

Library Management System

Project Statement & Comprehensive Specifications

Python 3.x

Console Application

Zero Dependencies

1. PROJECT OBJECTIVE

The primary objective of the Library Management System is to provide a simple, beginner-friendly, and lightweight console application written in pure Python to manage basic library operations. It aims to help small libraries or individual users track books, manage inventory copies, and handle book borrowing and returning processes without requiring external databases or third-party libraries.

2. TARGET AUDIENCE

Beginner Programmers

Individuals learning Python who want to study clean, modular procedural code, lists, and dictionary structures.

Small Libraries & Educators

Organizers needing a minimal offline tool, or instructors teaching basic programming logic and input validation.

3. OVERVIEW & TECHNICAL SPECIFICATIONS

The application operates as a console-based system built entirely in native Python 3. It utilizes volatile in-memory storage, meaning data resets upon program termination, ensuring clean isolation for testing and learning workflows.

- **Language:** Python 3.x
- **Dependencies:** None (uses standard built-in Python features and built-in data structures).
- **Data Persistence:** In-memory storage (lists and dictionaries).

4. FUNCTIONAL REQUIREMENTS

- **Add a Book:** Register new books into the system by specifying the title, author name, and number of total copies. Each book is automatically assigned a unique ID.
- **View All Books:** Display a formatted table listing all available books, including their ID, title, author, and current available copy count.
- **Search for a Book:** Search the library catalog using keywords matching part of a book's title or author name.
- **Issue a Book:** Record the issuance of a book to a borrower by providing the Book ID and borrower's name, provided that a copy is currently available and the borrower hasn't already checked out that specific book.
- **Return a Book:** Process the return of a book by a borrower, updating the active issuance record.
- **Delete a Book:** Remove a book from the system using its Book ID, ensuring that no copies are currently checked out before deletion.

5. HOW TO RUN

Execute the application locally via terminal or command prompt using the following steps:

- Ensure Python 3 is installed on your system environment.
- Save the application script into a file named `library_management.py`.
- Open a terminal or command prompt, navigate to the folder containing the script, and execute:

```
python "library_management.py"
```

Follow the interactive menu prompts displayed in the terminal to manage inventory seamlessly.

6. AUTHOR & MAINTENANCE

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